

## EDUCATION

**Kyungpook National University (KNU)**  
*Bachelor of Science in Physics*  
Overall GPA: 4.11/4.3 | Major GPA (71 credits): 4.19/4.3

Daegu, Republic of Korea  
Mar. 2020 – Feb. 2024

## PUBLICATIONS

[4] **Magnetic characterization of electronic components for portable atomic sensors using a zero-field optically pumped magnetometry platform**  
*Manuscript submitted; arXiv:2609.07270 (2026). DOI: 10.48550/arXiv:2609.07270*  
H. Kim, S. Lee, S. H. Yim, T. Jeong, and Y. Lim

[3] **Wire-sealed atomic gas cells for noble-gas nuclear magnetic resonance applications**  
*Manuscript under review*  
S. H. Hong, Y. Lim, S. H. Yim, J. B. Nam, S. Lee, T. Jeong, and H. Kim

[2] **Magnetic Anomaly Detection of an Undersea Tunnel Using a Drone-Mounted All-Optical Pulsed Atomic Magnetometer**  
*IEEE Sensors Journal*, Early Access (2026). DOI: 10.1109/JSEN.2026.3729190  
H. Kim, S. H. Yim, S. Lee, E.-S. Bang, S. R. Yeo, Y. J. Kim, T. Jeong, and Y. Lim

[1] **Aging test of atomic vapor cell with Al<sub>2</sub>O<sub>3</sub> wall coating on cubic glass**  
*Applied Optics* **64**, 7932–7937 (2025). DOI: 10.1364/AO.572036  
H. Kim, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim

## SELECTED PRESENTATIONS

[3] **Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer**  
*Poster presentation; APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting (2026)*

[2] **Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells**  
*Invited oral presentation; KISTI-SNU Joint Workshop (2026)*

[1] **Lifetime extension of rubidium vapor cells by Al<sub>2</sub>O<sub>3</sub> coating**  
*Poster presentation; APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting (2025)*

## RESEARCH EXPERIENCE

**Agency for Defense Development (ADD)**  
*Research Officer for National Defense (Military service)*  
Supervisors: Dr. Sin Hyuk Yim and Dr. Sangkyung Lee  
Research in experimental atomic physics and quantum sensing

Daejeon, Republic of Korea  
Jun. 2024 – Present

- Developed an automated absorption spectroscopy system with spectral modeling to characterize atomic vapor cells and contributed to their fabrication
- Built a zero-field optically pumped magnetometry platform to evaluate component-induced magnetic interference and provide design guidance for portable atomic sensors
- Deployed a drone-mounted atomic magnetometer for magnetic anomaly detection and established a quantitative framework for interpreting the acquired data

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| <b>Kyungpook National University (KNU)</b>                                                                                                                                                                                                                                                                                                                           | Daegu, Republic of Korea                       |
| <i>Undergraduate Researcher</i>                                                                                                                                                                                                                                                                                                                                      |                                                |
| <ul style="list-style-type: none"> <li><b>Novel Applied Nano Optics Lab</b> (Advisor: Prof. Junyeob Yeo)<br/>Fabricated a flexible photoelectrochemical cell using laser processing</li> <li><b>High Energy Physics Lab (Moon Lab)</b> (Advisor: Prof. Chang-Sung Moon)<br/>Developed momentum reconstruction algorithms for silicon detector simulations</li> </ul> | Jun. 2021 – Jun. 2022<br>Jun. 2020 – Feb. 2021 |

**HONORS AND AWARDS**

|                                                                                                                                                           |                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>KFAS Study Abroad Workshop (National Sciences)</b><br>Korea Foundation for Advanced Studies (KFAS)                                                     | Jul. 2026                       |
| <b>Physics Alumni Association Award and Scholarship (Outstanding Graduate)</b><br>Department of Physics Alumni Association, Kyungpook National University | Feb. 2024                       |
| <b>2nd Prize, 2023 MiliTECH Challenge</b><br>Hosted by KAIST; awarded by the Agency for Defense Development                                               | Dec. 2023                       |
| <b>2nd Prize, 2022 MiliTECH Challenge</b><br>Hosted by KAIST; awarded by the Korea Military Academy                                                       | Dec. 2022                       |
| <b>3rd Prize, Academic Conference</b><br>College of Natural Sciences, Kyungpook National University                                                       | Dec. 2021                       |
| <b>Dean’s List</b><br>College of Natural Sciences, Kyungpook National University                                                                          | Nov. 2021, Apr. 2022, Apr. 2023 |

**GRANTS AND SCHOLARSHIPS**

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|-------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>National Science and Technology Scholarship</b><br>Korea Student Aid Foundation (KOSAF)                  | Mar. 2022 – Feb. 2024           |
| <b>Academic Merit Scholarship (Hyoseok Scholarship)</b><br>Kyungpook National University Alumni Association | Apr. 2022                       |
| <b>Academic Excellence Scholarship</b><br>Kyungpook National University                                     | Aug. 2020, Feb. 2021, Aug. 2021 |

**OTHER RESEARCH OUTPUTS**

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|-------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Buffer Gas Pressure Estimation for OPMs</b><br>Registered software; Korea Copyright Commission (C-2024-047156)                   | Dec. 2024 |
| <b>Rubidium Vapor Cell Lifetime Monitoring and Analysis Tool</b><br>Registered software; Korea Copyright Commission (C-2024-054414) | Dec. 2024 |

**ADDITIONAL EXPERIENCE**

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|-------------------------------------------------------------|-----------------------|
| <b>Teaching Assistant, Classical Mechanics II (PHYS212)</b> | Sep. 2023 – Dec. 2023 |
| <b>Teaching Assistant, Electromagnetism II (PHYS312)</b>    | Mar. 2023 – Jun. 2023 |
| <b>International Student Tutor, KNU</b>                     | Feb. 2022 – May 2022  |

**SKILLS**

**Programming:** Python, Wolfram Language | **Software:** LabVIEW, SolidWorks, COMSOL Multiphysics